Control Theory

Chap 7: Kalman Filter

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November 12, 2026



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Chap 7: Kalman Filter

1. Standard Kalman Filter
2. Kalman - Bucy Filter
3. Hybrid Kalman Filter
4. Chapter Conclusion



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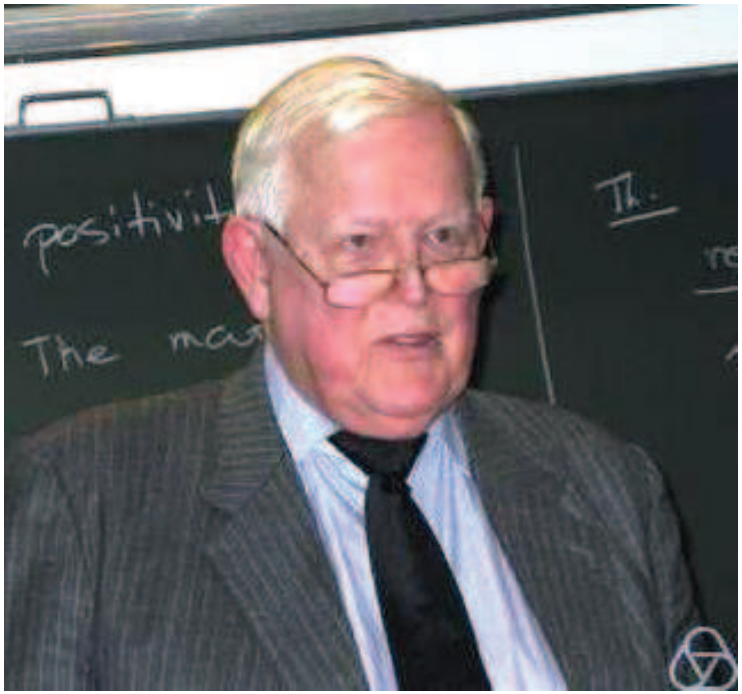


Figure 1: Prof. Rudolf E. Kálmán

Kalman filter

Also linear quadratic estimation (LQE).

- Considering statistical noise and inaccurate sensor bias
- Use a series of measurements over time and produce estimates of unknown variables.
- More accurate than those based on a single measurement alone.
- Estimate a joint probability distribution over the variables for each time-frame.

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The original discrete system

$$\begin{aligned} x_k &= Ax_{k-1} + Bu_k + \omega_k \\ z_k &= Hx_k + \nu_k \end{aligned} \tag{1}$$

$$\begin{aligned} \omega_k &\sim \mathcal{N}(0, Q), E(\omega_k^T \omega_k) = Q \\ \nu_k &\sim \mathcal{N}(0, R), E(\nu_k^T \nu_k) = R \end{aligned}$$

ω_k and ν_k are assumed independent, white, and with normal probability. Q and R are the covariance matrices.

The Kalman filter can be stated as

$$\begin{aligned} \hat{x}_k^{'} &= A\hat{x}_{k-1} + Bu_{k-1} \\ P_k^{'} &= AP_{k-1}A^T + Q \end{aligned} \tag{2}$$

where the update equations are

$$\begin{aligned} K_k^{'} &= P_k^{'} H^T \left(HP_k^{'} H^T + R \right)^{-1} \\ \hat{x}_k &= \hat{x}_k^{'} + K_k(z_k - H\hat{x}_k^{'}) \\ P_k &= (I - K_k H)P_k^{'} \end{aligned} \tag{3}$$

$\hat{x}_k$: the estimate of x_k.

Proof.

A priori state estimate error e_k^- and a posteriori estimate error e_k are

$$\begin{aligned} e_k^- &= x_k - \hat{x}_k^- \\ e_k &= x_k - \hat{x}_k \end{aligned} \tag{4}$$

where $\hat{x}_k^-$ and $\hat{x}_k$ are the priori and posteriori state estimates of real system state x_k. A priori estimate error covariance is

$$P_k^- = E[e_k^- e_k^{-T}]$$

and a posteriori estimate error covariance is

$$P_k = E[e_k e_k^T] \tag{5}$$

The posteriori state estimate is calculated based on the priori estimate as

$$\hat{x}_k = \hat{x}_k^- + K(z_k - H\hat{x}_k^-) \tag{6}$$

□

Proof.

$z_k - H\hat{x}_k^-$ is the innovation(or residual), reflecting the discrepancy between the predicted measurement $H\hat{x}_k^-$ and the actual measurement z_k.

If $z_k - H\hat{x}_k^-$ is zero, indicating measurement z_k matches the gained estimate $H\hat{x}_k^-$ well, and the priori state $\hat{x}_k^-$ matches the posteriori state $\hat{x}_k$ accurately.

K is called the *gain* or *blendingfactor* that minimizing (5), which is also the optimization target.

$$\min P_k \tag{7}$$

□

Proof.

As defined above, one can verify that

$$
\begin{aligned}
P_k &= \mathbf{cov}(e_k) = \mathbf{cov}(x_k - \hat{x}_k) \\
&= \mathbf{cov}(x_k - \hat{x}_k^- - K(z_k - H\hat{x}_k^-)) \\
&= \mathbf{cov}(x_k - \hat{x}_k^- - K(Hx_k + \nu_k - H\hat{x}_k^-)) \\
&= \mathbf{cov}(x_k - \hat{x}_k^- - KHx_k - K\nu_k + KH\hat{x}_k^-) \\
&= \mathbf{cov}(x_k - KHx_k - \hat{x}_k^- + KH\hat{x}_k^- - K\nu_k) \\
&= \mathbf{cov}\ ((I - KH)(x_k - \hat{x}_k^-) - K\nu_k) \\
&= \mathbf{cov}\ ((I - KH)(x_k - \hat{x}_k^-) - K\nu_k) \\
&= (I - KH)\mathbf{cov}\ (x_k - \hat{x}_k^-)(I - KH)^T + KRK^T \\
&= (I - KH)\mathbf{cov}\ (e_k^-)(I - KH)^T + KRK^T \\
&= (I - KH)P_k^-(I - KH)^T + KRK^T
\end{aligned}
\tag{8}
$$

The posteriori estimate error

$$e_k^- = x_k - \hat{x}_k$$

□

Proof.

The minimization of P_k, i.e. $E[|x_k - \hat{x}_k|^2]$, which equals to the minimization trace of covariance. From (8), it has the extension that

$$\begin{aligned} P_k &= (I - KH)P_k^-(I - KH)^T + KRK^T \\ &= P_k^- - KHP_k^- - P_k^- H^T K^T + KHP_k^- H^T K^T + KRK^T \\ &= P_k^- - KHP_k^- - P_k^- H^T K^T + KSK^T \end{aligned} \tag{9}$$

where $S = HP_k^- H^T + R$. By the derivative of matrix, it has the minimization of trace, which it has

$$\frac{dtr(P_k)}{dK} = -2(HP_k^-)^T + 2KS = 0 \tag{10}$$

then it can obtain the optimal gain K that

$$K = (HP_k^-)^T S^{-1} = (HP_k^-)^T (HP_k^- H^T + R)^{-1} \tag{11}$$



Kalman filter reexpression

$$\begin{aligned} \hat{x}_k &= \hat{x}_k^{'} + K_k \tilde{y}_k \\ \tilde{y}_k &= z_k - H_k \hat{x}_k \end{aligned} \tag{12}$$

where

$$\begin{aligned} K_k &= P_k^{'} H_k^T S_k^{-1} \\ S_k &= H_k P_k H_k^T + R \\ P_k &= (I - K_k H_k) P_k \\ \hat{x}^{'} &= \Phi \hat{x}_k \\ P_{k+1} &= \Phi P_k \Phi^T + Q \end{aligned}$$

Kalman filter brief expression

$$\hat{x}_{k+1} = \Phi x_k + B u_k + K_k (z_k - H \hat{x}_k) \tag{13}$$

where

$$\begin{aligned} K_k &= P_k^{'} H^T \left(H P_k^{'} H^T + R \right)^{-1} \\ P_k &= (I - K_k H) P_k^{'} \\ P_{k+1}^{'} &= \Phi P_k \Phi^T + Q \end{aligned}$$

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Consider the state-space model

$$\begin{cases} \dot{x}(t) &= A(t)x(t) + B(t)u(t) + w(t) \\ z(t) &= H(t)x(t) + \nu(t) \end{cases} \tag{14}$$

where

$$\begin{aligned} w(t) &\sim \mathcal{N}(0, Q(t)) \\ \nu(t) &\sim \mathcal{N}(0, R(t)) \end{aligned}$$

Continuous Kalman Filter(Observer)

Then the filter can be constructed as

$$\begin{cases} \dot{\hat{x}}(t) &= A(t)\hat{x}(t) + B(t)u(t) + K(t)(z(t) - H(t)\hat{x}(t)) \\ \dot{P}(t) &= A_{kh}(t)P(t) + P(t)A_{kh}^T(t) + Q(t) + K(t)R(T)K^T(t) \end{cases} \tag{15}$$

the Kalman Gain is calculated as

$$K(t) = P(t)H^T(t)R^{-1}(t) \tag{16}$$

Proof.

The filter error dynamics is

$$\begin{aligned} \dot{e}(t) &= A(t)e(t) + w(t) - K(t)\left(z(t) - H(t)\hat{x}(t)\right) \\ &= \left(A(t) - K(t)H(t)\right)e(t) + w(t) - K(t)\nu(t) \\ &= A_{kh}e(t) + w(t) - K(t)\nu(t) \end{aligned} \tag{17}$$

where $A_{kh} = A(t) - K(t)H(t)$. Set the optimization target

$$J = E\left(e^T(t)e(t)\right) \tag{18}$$

Define the covariance matrix

$$P\left(e(t), e(t)\right) = E\left([x(t) - \hat{x}(t)][x(t) - \hat{x}(t)]^T\right) \in \mathcal{R}^{n\times n} \tag{19}$$

which is simplified expressed as $P(t)$, with (17) one has that

$$\begin{aligned} \dot{P}(t) &= E\left(\dot{e}(t)e^T(t)\right) + E\left(e(t)\dot{e}^T(t)\right) \\ &= E\left((A_{kh}e(t) + w(t) - K(t)\nu(t))e^T(t)\right) + E\left(e(t)(A_{kh}e(t) + w(t) - K(t)\nu(t))^T\right) \\ &= A_{kh}P(t) + P(t)A_{kh}^T + E(w(t)w^T(t)) + E([K(t)\nu(t)][K(t)\nu(t)]^T) \\ &= A_{kh}P(t) + P(t)A_{kh}^T + Q(t) + K(t)R(t)K^T(t) \end{aligned} \tag{20}$$

Proof.

According to the minimum - maximum theorem, one needs to choose gain K so as to minimize $tr[\dot{P}]$,i.e.

$$H = \textbf{min}\left[\dot{P}\right] \tag{21}$$

Differentiate with respect to K and equate to zero

$$\frac{\partial H}{\partial K} = -2PH + 2KR = 0 \tag{22}$$

it has that

$$K(t) = PH^T R^{-1} \tag{23}$$

Submit (23) into (20), one has that

$$\begin{aligned} \dot{P} &= A_{kh}P(t) + P(t)A_{kh}^T + Q(t) + K(t)R(t)K^T(t) \\ &= A_{kh}P(t) + P(t)A_{kh}^T + Q(t) + PH^T R^{-1} HP^{-T} \\ &= AP(t) + P(t)A^T + Q(t) - PH^T R^{-1} HP \end{aligned} \tag{24}$$

□

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The continuous system model and the discrete measurement output are

$$\begin{cases} \dot{x}(t) &= A(t)x(t) + B(t)u(t) + w(t) \\ z_k &= H_k x_k + \nu_k \end{cases} \tag{25}$$

where

$$x_k = x(t_k), w(t) \sim \mathcal{N}(0, Q(t)), \nu_k \sim \mathcal{N}(0, R_k)$$

With the time-varying sampling periods T_k, it has that

$$\begin{cases} x_k &= F_k x_{k-1} + G_k u_k + \Lambda w_k \\ y_k &= H_k x_k + \nu_k \end{cases} \tag{26}$$

where $T_k = t_k - t_{k-1}$, with $k = 1, ..., +\infty$, t_k is the $k-th$ sampling instant, $x_k = x(t_k)$.

$$\begin{array}{rclcrcl} F_k &=& exp(AT_k) & \approx & & (I + AT_k + \frac{1}{2}(AT_k)^2) \\ G &=& \left(e^{AT_k} - I\right) A^{-1} B & \Lambda &=& \left(e^{AT_k} - I\right) A^{-1} \\ H &=& C & Q_k &\sim& QT \\ R_k &\sim& \frac{R}{T} & & & \end{array}$$



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- Optimization Controller - Optimization Observer - Optimization Filter
- Discrete System - Continuous System - Hybrid System
- Filter - Observer - Estimator



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谢谢!

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